# Vector Fields and Line Integrals

Lecture 02

Instructor: Asgher Ali

September 11th, 2025

# Flow Integrals and Circulation for Velocity Fields

- Suppose that $\mathbf{F}$ represents the velocity field of a fluid flowing through a region in space (a tidal basin or the turbine chamber of a hydroelectric generator, for example).
- Under these circumstances, the integral of $\mathbf{F} \cdot \mathbf{T}$ along a curve in the region gives the fluid flow ***along*** or ***circulation*** around, the curve.

**Definition**

If $\mathbf{r(t)}$ parametrizes a smooth curve $C$ in the domain of a continuous velocity field $\mathbf{F}$, the **flow** along the curve from $A = \mathbf{r}(a)$ to $B = \mathbf{r}(b)$ is

$$\text{Flow} = \int_C (\mathbf{F} \cdot \mathbf{T}) ds$$

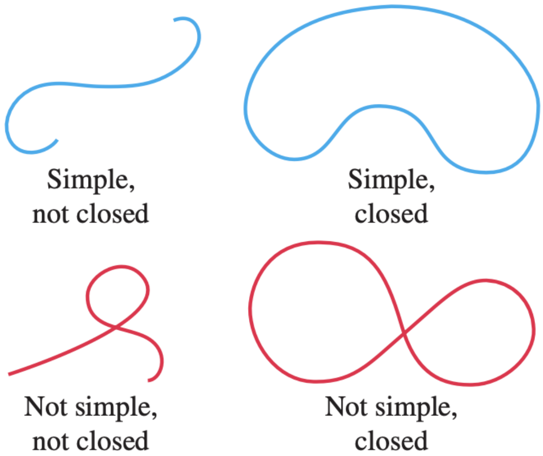
- The integral is called a **flow integral**.
- If the curve starts and ends at the same point, so that $A = B$, the flow is called the **circulation**.

**Examples**

- A fluid's velocity field is $\mathbf{F} = x\mathbf{i} + z\mathbf{j} + y\mathbf{k}$. Find the flow along the helix $\mathbf{r}(t) = \cos(t)\mathbf{i} + \sin(t)\mathbf{j} + t\mathbf{k}$, $0 \leq t \leq \frac{\pi}{2}$.
- Find the circulation of the field $\mathbf{F} = (x - y)\mathbf{i} + x\mathbf{j}$ around the circle $\mathbf{r}(t) = \cos(t)\mathbf{i} + \sin(t)\mathbf{j}$, $0 \leq t \leq 2\pi$.

## Flux across a Simple Closed Plane Curve

- A curve in the $xy$-plane is **simple** if it does not cross itself.
- When a curve starts and ends at the same point, it is a **closed curve** or **loop**.




- To find the rate at which fluid is entering or leaving a region enclosed by a smooth simple closed curve $C$ in the plane, we calculate the line integral over $C$ of $\mathbf{F} \cdot \mathbf{n}$, the scalar component of the fluid's velocity field in the direction of the curve's outward-pointing normal vector.
- The difference between flux and circulation is that flux of $F$ across $C$ is the line integral with respect to the arc length of $\mathbf{F} \cdot \mathbf{n}$, the scalar component of $\mathbf{F}$ in the direction of **outward-normal**, whereas the circulation of $\mathbf{F}$ around $C$ is the line integral of $\mathbf{F} \cdot \mathbf{T}$, scalar component of $\mathbf{F}$ in the direction of **unit tangent vector**.

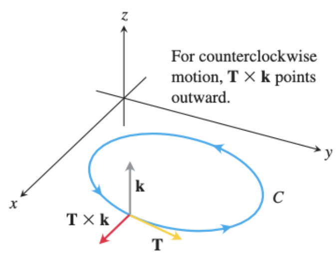
## Definition

If $C$ is a smooth simple closed curve in the domain of a continuous vector field $\mathbf{F} = M(x,y)\mathbf{i} + N(x,y)\mathbf{j}$ in the plane and if $\mathbf{n}$ is the outward-pointing normal vector on $C$, the **flux** of $\mathbf{F}$ across $C$ is

$$\text{Flux of } \mathbf{F} \text{ across } C = \int_C (\mathbf{F} \cdot n)\, ds$$

## Calculating Flux using a simple computation

Instead of computing $\mathbf{n}$ and $\mathbf{ds}$, we formulate by considering




In terms of components,

$$\begin{aligned}
\mathbf{n} &= \mathbf{T} \times k \\
&= \left(\frac{dx}{ds}\mathbf{i} + \frac{dy}{ds}\mathbf{j}\right) \times \mathbf{k} \\
&= \frac{dy}{ds}\mathbf{i} - \frac{dx}{ds}\mathbf{j}
\end{aligned}$$

If $\mathbf{F} = M(x,y)\mathbf{i} + N(x,y)\mathbf{j}$, then

$$\begin{aligned}
\int_C (\mathbf{F} \cdot n)\, ds &= \int_C \left(M\frac{dy}{ds} - N\frac{dx}{ds}\right) ds \\
&= \oint M dy - N dx
\end{aligned}$$

The integral can be evaluated from any smooth parametrization $x = g(t), y = h(t)$, $a \leq t \leq b$, that traces $C$ counterclockwise exactly once.

$$\boxed{\text{Flux of } \mathbf{F} \text{ across } C = \oint M dy - N dx}$$

We put $\circ$ on the integral as a reminder that the integration around the closed curve $C$ is to be in the counterclockwise direction.

**Example**

Find the flux of $\mathbf{F} = (x-y)\mathbf{i} + x\mathbf{j}$ across the circle $x^2+y^2=1$ in the $xy-$plane.

The parametrization $\mathbf{r}(t) = \cos(t)\mathbf{i} + \sin(t)\mathbf{j}$, $0 \le t \le 2\pi$, traces the circle counterclockwise exactly once. We can therefore use this parametrization with

$$\begin{aligned} M &= x - y = \cos(t) - \sin(t) \\ N &= x = \cos(t) \end{aligned}$$

and

$$\begin{aligned} dy &= d(\sin(t)) = \cos(t)dt \\ dx &= d(\cos(t)) = -\sin(t)dt \end{aligned}$$

Now,

$$\begin{aligned} \text{Flux} &= \oint M dy - N dx \\ &= \int_0^{2\pi} \Big(\cos^2(t) - \sin(t)\cos(t) + \cos(t)\sin(t)\Big)\, dt \\ &= \int_0^{2\pi} \cos^2(t) dt \\ &= \pi \end{aligned}$$

## Example

Find the circulation and flux of the field $\mathbf{F}$ around and across the closed semicircular path that consists of the semicircular arch $\mathbf{r}_1(t) = (a\cos(t))\mathbf{i} + (a\sin(t))\mathbf{j}$, $0 \leq t \leq \pi$, followed by the line segment $\mathbf{r}_2(t) = t\mathbf{i}$, $-a \leq t \leq a$.

- $\mathbf{F} = x\mathbf{i} + y\mathbf{j}$
- $\mathbf{F} = -y\mathbf{i} + x\mathbf{j}$
- $\mathbf{F} = x^2\mathbf{i} + y^2\mathbf{j}$
- $\mathbf{F} = -y^2\mathbf{i} + x^2\mathbf{j}$

## Example

Find the outward flux of the field $\mathbf{F} = (x+y)\mathbf{i} - (x^2+y^2)\mathbf{j}$ across the

- triangle with vertices $(1,0), (0,1)$ and $(-1,0)$